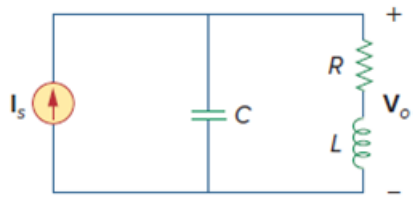


Problem

For the circuit shown in Fig. 14.72, find $\mathbf{H}(s) = \mathbf{V}_o/\mathbf{I}_s$.

Figure 14.72:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 14.72 in the text book.

In frequency domain inductor is represented as $j\omega L$, capacitor is represented as $\frac{1}{j\omega C}$.

The frequency domain circuit is shown in Figure 1.

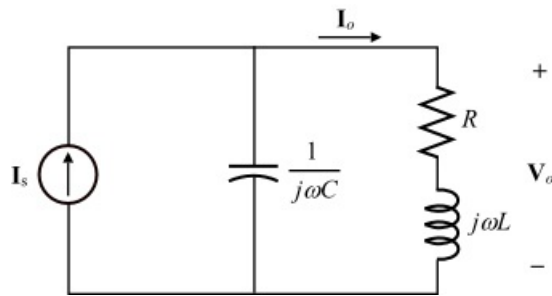


Figure 1

Step 2 of 3

Using the current division rule, calculate the current through the resistor and inductor branch.

$$\begin{aligned} \mathbf{I}_o &= \left(\frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R + j\omega L} \right) \mathbf{I}_s \\ &= \left(\frac{\frac{1}{j\omega C}}{\frac{1 + j\omega RC + (j\omega L)(j\omega C)}{j\omega C}} \right) \mathbf{I}_s \\ &= \left(\frac{1}{1 + j\omega RC - \omega^2 LC} \right) \mathbf{I}_s \end{aligned}$$

Step 3 of 3

Calculate the transfer function.

$$\mathbf{V}_o = (R + j\omega L)\mathbf{I}_o$$

Substitute $\left(\frac{1}{1 + j\omega RC - \omega^2 LC}\right)\mathbf{I}_s$ for $\mathbf{I}_o$.

$$\mathbf{V}_o = (R + j\omega L)\left(\frac{1}{1 + j\omega RC - \omega^2 LC}\right)\mathbf{I}_s$$

$$\frac{\mathbf{V}_o}{\mathbf{I}_s} = \frac{R + j\omega L}{1 + j\omega RC - \omega^2 LC}$$

$$\mathbf{H}(j\omega) = \frac{R + j\omega L}{1 + R(j\omega)C + (j\omega)^2 LC}$$

Substitute s for $j\omega$ in the above equation.

$$\begin{aligned}\mathbf{H}(s) &= \frac{R + sL}{1 + R(sC) + s^2 LC} \\ &= \frac{sL + R}{s^2 LC + sRC + 1}\end{aligned}$$

Therefore, the transfer function, $\mathbf{H}(s)$ is $\boxed{\frac{sL + R}{s^2 LC + sRC + 1}}$.